18) Experiment to understand Matplotlib library use cases in Data Science through visualization.

Program:

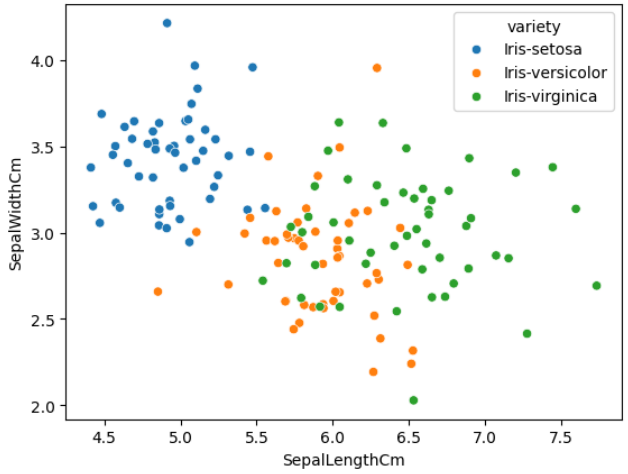
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
np.random.seed(42)
data = pd.DataFrame({
    'SepalLengthCm': np.concatenate([
        np.random.normal(5.0, 0.3, 50),
        np.random.normal(5.9, 0.4, 50),
        np.random.normal(6.5, 0.5, 50)
    ]),
    'SepalWidthCm': np.concatenate([
        np.random.normal(3.4, 0.3, 50),
        np.random.normal(2.8, 0.3, 50),
        np.random.normal(3.0, 0.3, 50)
    ]),
    'PetalLengthCm': np.concatenate([
        np.random.normal(1.5, 0.2, 50),
        np.random.normal(4.3, 0.4, 50),
        np.random.normal(5.5, 0.5, 50)
    ]),
    'PetalWidthCm': np.concatenate([
        np.random.normal(0.2, 0.1, 50),
        np.random.normal(1.3, 0.2, 50),
        np.random.normal(2.0, 0.3, 50)
```

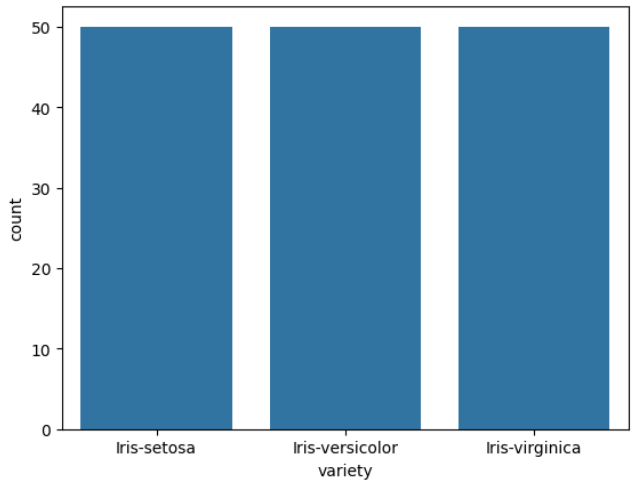
```
    ]),
    'variety': ['Iris-setosa'] * 50 +
               ['Iris-versicolor'] * 50 +
               ['Iris-virginica'] * 50
})
data['Id'] = range(1, 151)
print(data.head())
print(data.info())
print(data.describe())
print(data.value_counts('variety'))
sns.countplot(x='variety', data=data)
plt.show()
sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm',
                hue='variety', data=data)
plt.show()
sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm',
                hue='variety', data=data)
plt.show()
sns.pairplot(data, hue='variety', height=3)
plt.show()
for column in ['PetalLengthCm', 'PetalWidthCm', 'SepalLengthCm']:
    sns.FacetGrid(data, hue='variety', height=5).map(
        sns.histplot, column
    ).add_legend()
    plt.show()
```

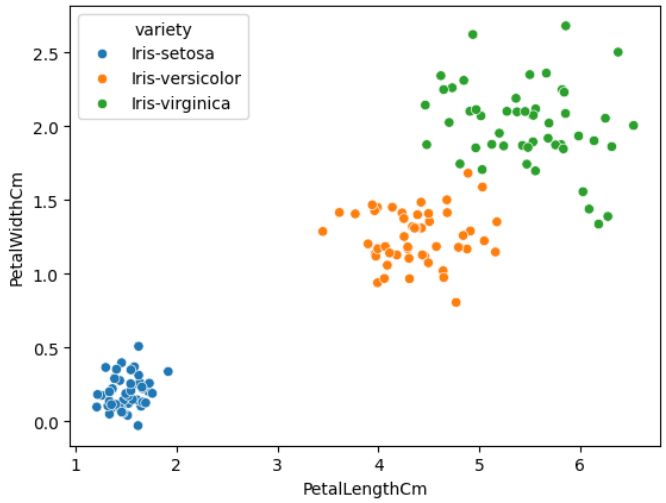
Output:

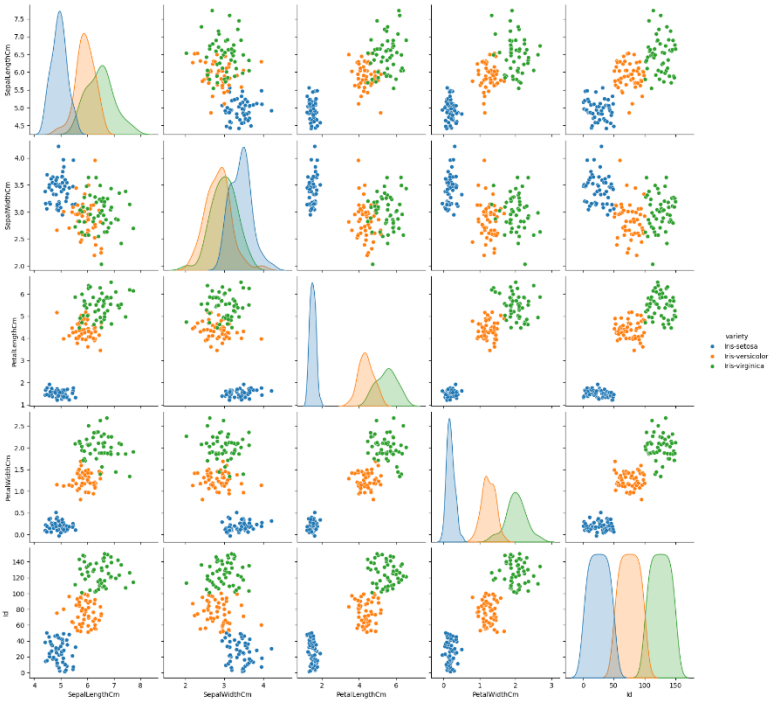
```
   SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      variety  Id
0       5.149014      3.475148       1.334201      0.193732  Iris-setosa   1
1       4.958521      3.503934       1.387964      0.295514  Iris-setosa   2
2       5.194307      3.195993       1.649459      0.101427  Iris-setosa   3
3       5.456909      3.469676       1.622074      0.250405  Iris-setosa   4
4       4.929754      3.487922       1.495820      0.146974  Iris-setosa   5
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   SepalLengthCm  150 non-null    float64
 1   SepalWidthCm   150 non-null    float64
 2   PetalLengthCm  150 non-null    float64
 3   PetalWidthCm   150 non-null    float64
 4   variety        150 non-null    object
 5   Id             150 non-null    int64
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
None
       SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm          Id
count     150.000000    150.000000     150.000000    150.000000  150.000000
mean        5.773275      3.088036       3.784888      1.150117   75.500000
std         0.749582      0.392937       1.723460      0.772978   43.445368
min         4.412099      2.027620       1.210383     -0.030192    1.000000
25%         5.096477      2.820749       1.625035      0.258596   38.250000
50%         5.809148      3.081515       4.349614      1.237650   75.500000
75%         6.298614      3.379593       5.106935      1.865582  112.750000
max         7.731621      4.216051       6.530374      2.681208  150.000000
```

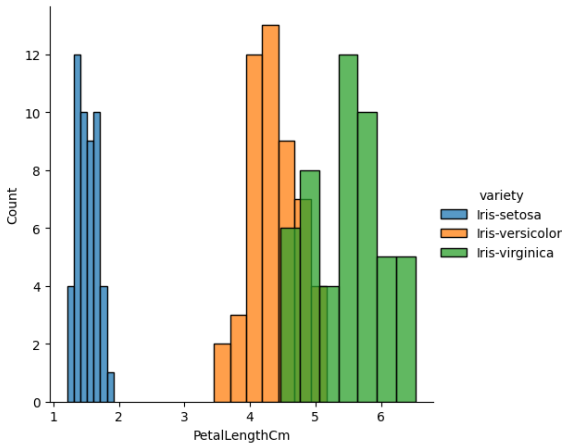
```
variety
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

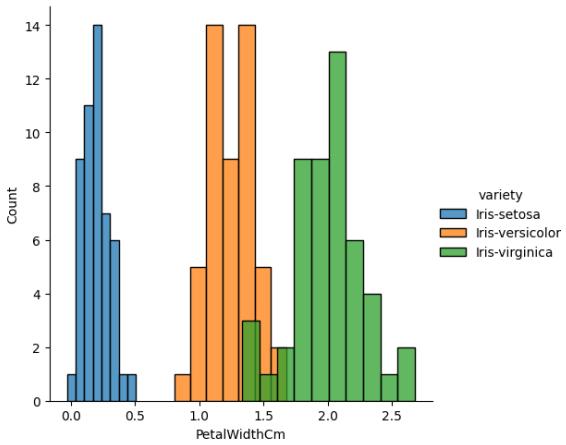


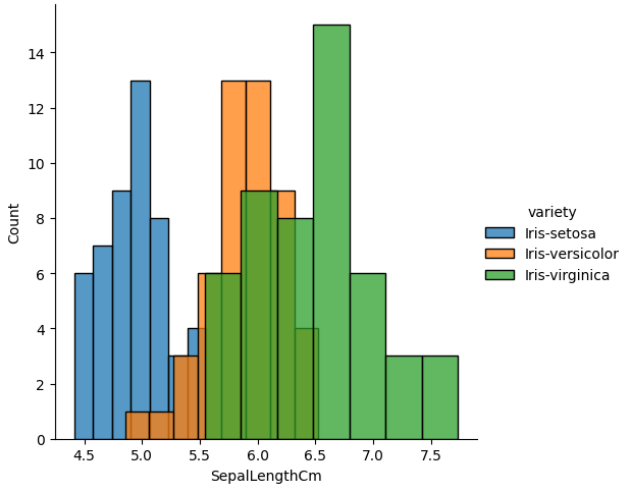



variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm


SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
Id
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalLengthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica


Count
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica



Result:

Thus, the Matplotlib and Seaborn libraries were used successfully to visualize the dataset using count plots, scatter plots, pairplots, and histograms.